



JAIPUR NATIONAL UNIVERSITY

Semester/Year End/Back Examination: December-2023

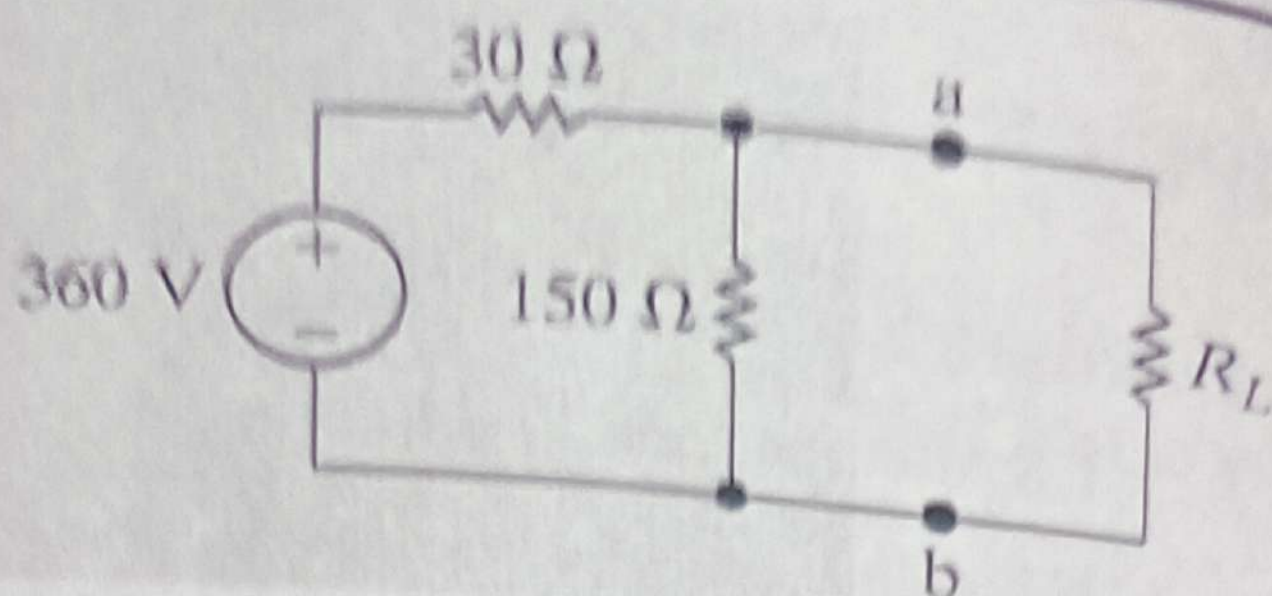
Name of School:	Engineering and Technology
Name of Programme / Course:	B.Tech. CSE(Hons.) in Artificial Intelligence / B.Tech (Biomedical) / B.Tech (Biotechnology) / B.Tech (Chemical) / B.Tech (Civil Engineering) / B.Tech (Computer Science) / B.Tech (Cyber Security) / B.Tech (Data Science) / B.Tech (Electrical Engineering) / B.Tech (Food Technology) / B.Tech (Mechanical)
Semester/Year:	I
Paper Code:	BTEEEDS101T22
Title of Paper:	Fundamental of Electrical and Electronics
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

Attempt any five Questions from section A and attempt any four Questions from section B.

Question Number	SECTION-A	Marks Allotted
Q.1	Define the Following Terms: - Current - Voltage - EMF - Power	4
Q.2	State and explain the Superposition theorem with suitable example.	4
Q.3	Deduce an expression for Single- phase A.C. circuits with RL Load.	4

Q.4



Find the value of R_L for maximum power transfer to R_L .

Q.5

Discuss the working principle of transformer with the help of suitable diagram.

Q.6

Define the concept of Transducer in brief.

Q.7

What do you understand by "Modulation"? Give Brief Description of various Modulations.

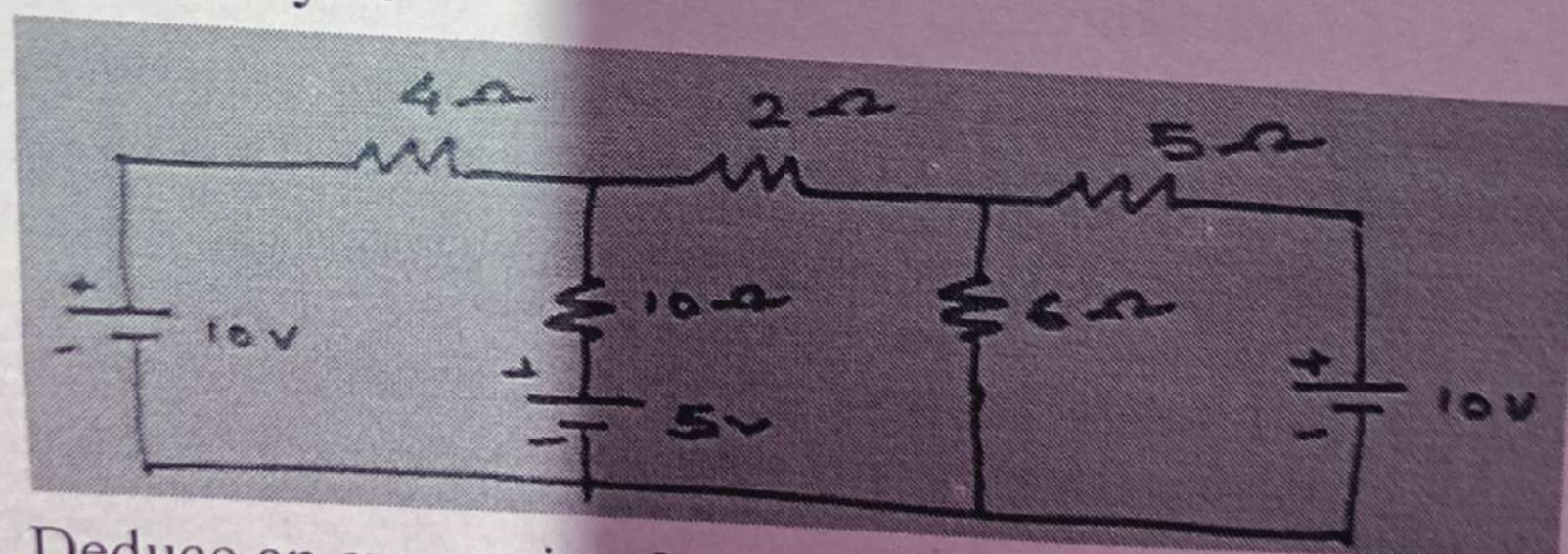
SECTION-B

Q.1

Deduce an expression for calculation of maximum power transfer with the help of thevenin's equivalent circuit.

Q.2

Calculate the branch currents in the given network with mesh analysis



Q.3

Deduce an expression for root mean square value of sinusoidal AC waveform with relevant waveform.

Q.4

Deduce an expression for EMF Equation of Transformer

Q.5

Draw and explain the V-I characteristics of PN Junction diode and Zener Diode.

Q.6

Explain the working principle of RTD with the help of neat circuit diagram.